AJAY ALLE

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SUMMARY

Aspiring Entry-level Data/AI-ML engineer with hands-on experience in Python, computer vision (OpenCV, MediaPipe) and embedded/IoT systems. Completed a Python with AI internship and an industrial automation internship, and built a real-time **Gesture-Controlled Smart Home Automation System** and an **Wi-Fi Based Attendance System Using MAC Address Detection**, delivering end-to-end solutions across software, hardware and data pipelines.

INTERNSHIPS

Python with AI Intern — EWB Edutech Pvt. Ltd. (Sep 2026 – Oct 2026) — *Python, NumPy, Pandas, CSV, Artificial Intelligence*

- Developed a foundation in writing structured python programs.
- Practised organized data with built-in collections and reusable functions.
- Learned object-oriented programming principles and class-based design.
- Gained exposure to numerical computation and tabular data analysis.
- Studied introductory neural-network mathematics and activation functions.
- Practised preparing a python projects for testing, documentation, and GitHub submission.

PROJECTS

Gesture-Controlled Smart Home Automation System (2026) — *Python, OpenCV, MediaPipe, Raspberry Pi, Arduino, Serial Communication*

- Developed an internet-free, gesture-controlled home automation system, eliminating Wi-Fi and cloud dependency.
- Owned the computer-vision pipeline built real-time hand-gesture detection with OpenCV and MediaPipe on Raspberry Pi, mapping finger-count patterns to appliance commands sent to an Arduino over serial communication to drive relay-based switching.
- Designed triple-layer control (gesture, IR remote, manual switch) with priority logic for safe manual/IR override; tested detection accuracy across varied lighting and hand positions.

Wi-Fi Based Attendance System Using MAC Address Detection (2025) — *ESP32, Embedded C, Google Apps Script, Google Sheets*

- Automated attendance tracking by detecting connected devices' MAC addresses over Wi-Fi on an ESP32 microcontroller, removing manual roll call.
- Built a Google Apps Script pipeline logging detections to Google Sheets, auto-calculating attendance percentage and flagging low-attendance students.
- Added an LCD and button interface for subject/semester selection with real-time multi-device support, delivering a scalable, paperless solution that reduced manual errors.

TECHNICAL SKILLS

Languages: Python, C, C++(basic)
Computer Vision & Embedded: OpenCV, MediaPipe, ESP32, Arduino, IoT concepts, sensor & serial interfacing
Industrial Automation: Programming, VFD operation, control panel wiring, motor control systems, troubleshooting & maintenance
AI & Python: Python-based AI development (EWB internship)
Tools: Jupyter Notebook, Git basics, Google Apps Script

CERTIFICATIONS

- **EWB EDU TECH PVT. LTD.** — PYTHON WITH AI [CERTIFICATE]
- **IIRS** — AI/ML GEODATA ANALYTICS [CERTIFICATE]
- **Microsoft** — Azure Cloud, Data & AI [CERTIFICATE]
- **Edunet Skills4Future** — Foundation Course on Artificial Intelligence and Green Skills [CERTIFICATE]
- **NPTEL (Swayam)** — The Joy of Computing Using Python [CERTIFICATE]
- **Outskill** — Generative AI Mastermind [CERTIFICATE]
- **Bootcamp** — Drone Technology: Foundation and Applications [CERTIFICATE]

EDUCATION

Dr. S. & S. S. Ghandhy Government Engineering College 2024 – 2027
Bachelor of Engineering (B.E.) in Electronics & Communication Engineering | Current CGPA: 7.52

Dr. S. & S. S. Ghandhy College of Engineering & Technology 2020 – 2023
Diploma in Electrical Engineering | CGPA: 6.90